Statement of Interest

CGCCR Research Fellow – UT MD Anderson Cancer Center — Yichao Jin, Ph.D.

Research Background

I received my Ph.D. in Public Policy and Political Economy from UT Dallas in May 2026, where my dissertation used a Discrete Choice Experiment (N=1,027) to measure how waiting time, treatment efficacy, side-effect burden, and financial incentives jointly shape health decisions. My methodological training spans structural choice modeling (mixed logit, latent class models), causal inference, and LLM-based behavioral simulation.

My core contribution to date is the Behavioral Digital Twin (BDT) framework, which anchors LLM-generated synthetic agents to empirically estimated choice frontiers from DCE data. Rather than treating LLM outputs as unconstrained predictions, the BDT framework uses a convex anchoring rule to enforce preference-consistent behavior, reducing monotonicity violations from 62.5% to 4.2% in vaccination decision modeling. While the application was vaccination, the underlying architecture—estimating patient preferences from choice data and using them as behavioral constraints in generative simulation—transfers directly to any domain where understanding and predicting patient trade-offs is the central question.

Why the Center for Goal Concordant Care Research

The CGCCR's mission—understanding what matters to patients with advanced cancer and aligning care accordingly—raises a methodological challenge that my work was built to address: how do you systematically measure what patients value when the stakes are highest, and how do you use that evidence to simulate how they would respond to different care pathways before those pathways are implemented?

What I bring is not just familiarity with DCE methods, but a proven ability to use choice data to construct behavioral models that generate testable predictions. My dissertation demonstrated that patients with high vs. low institutional trust respond to waiting time in fundamentally different ways—a finding with direct implications for how advanced cancer patients with different trust profiles might respond to treatment delay, care transitions, or communication strategies. I am interested in the Center's research not as a new direction, but as the clinical setting where my methodological toolkit finds its most natural and consequential application.

Future Research at CGCCR

If appointed, I would pursue three interconnected lines of research:

1. Preference heterogeneity in goal-concordant care. Advanced cancer patients facing treatment decisions weigh efficacy, side effects, quality of life, and care burden simultaneously. I would apply latent class and mixed logit models to DCE data from CGCCR studies to identify distinct preference classes among advanced cancer patients and map them onto clinical trajectories. The goal is to move beyond one-size-fits-all communication and decision-support toward preference-stratified approaches.

2. Simulation-based pre-testing of communication and decision-support interventions. The BDT framework can be adapted to simulate how different patient segments would respond to alternative shared decision-making tools, goal-concordance conversations, or care coordination models before they are field-tested. This shifts the Center's intervention pipeline from trial-and-error toward simulation-informed design.

3. Preference dynamics over the care trajectory. Patient goals shift as cancer progresses. I would develop longitudinal preference measurement approaches—repeated DCE modules embedded in clinical follow-up—to track how treatment priorities evolve and identify transition points where care is most likely to lose alignment with patient goals.

These lines of research are designed to sit within the CGCCR's existing strengths in communication science, implementation science, and supportive care. My role would be to provide the quantitative preference measurement and behavioral simulation infrastructure that connects these disciplines to measurable patient outcomes.

Publications

1. "Empirical-Frontier Regularization for LLM Synthetic Agents: A Preference-Anchored Framework" — *Working Paper, target: Journal of Biomedical Informatics*. Presented at AIE Summer Conference 2026 and DAISY 2026 (ACM CHIL 2026).
2. "Value Contamination in Policy Simulation: Measuring How LLM Policy-Value Orientations Shape Predicted Administrative Burden Effects" — *Working Paper*. An audit framework testing nine LLMs across six model families (N=2,160 observations).
3. "Waiting Time as a Behavioral Barrier to Vaccination Uptake: Nonlinear Heterogeneity by Institutional Trust" — *Under Review at Economic Analysis and Policy*.
4. "The Price of Waiting: Evidence on Cash–Time Trade-Offs in Vaccination Time Discount" — *Manuscript in preparation, expected submission August 2026*.

Career Goals

My long-term goal is to lead an independent research program at the intersection of patient preference measurement and computational behavioral modeling, with a focus on translating preference evidence into clinical tools that improve decision quality for patients facing serious illness. The CGCCR fellowship offers the clinical context, multidisciplinary mentorship, and patient-centered research infrastructure to develop the pilot data and publication record needed to compete for early career awards (K01, K99/R00) and ultimately establish a faculty research program in preference-sensitive decision support.